

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: LSTM
group_sec: 80
timesteps_orig: 60
batch_size: 256
min_units: 16
max_units: 1024
units_step: 16
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
group_msec: 80
samples_per_sec: 12
forecast_horizon_sec: 4
forecast_steps: 48
timesteps: 60

LSTM Multi-Step Training Report

Report Generated: 2025-05-20 23:08:17

Training Start: 2025-05-20 21:46:17

Training End: 2025-05-20 22:58:55

Training Duration: 72.64 minutes

Predictions Start: 2025-05-20 22:58:55

Predictions End: 2025-05-20 23:00:37

Predictions Duration: 1.69 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: elu

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 1.7203

MSE: 4.8758

RMSE: 2.2081

R2 Score: 0.7066

Multi-Step Forecast:

Forecast Steps: 48

MAE: 0.0188

MSE: 0.0006

RMSE: 0.0238

R2 Score: 0.9386

Dataset size before down-sample: 85,269 rows
Dataset size after down-sample: 12,829 rows

First 5 timestamps BEFORE:

1744397085760
1744397085770
1744397085789
1744397085801
1744397085817

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

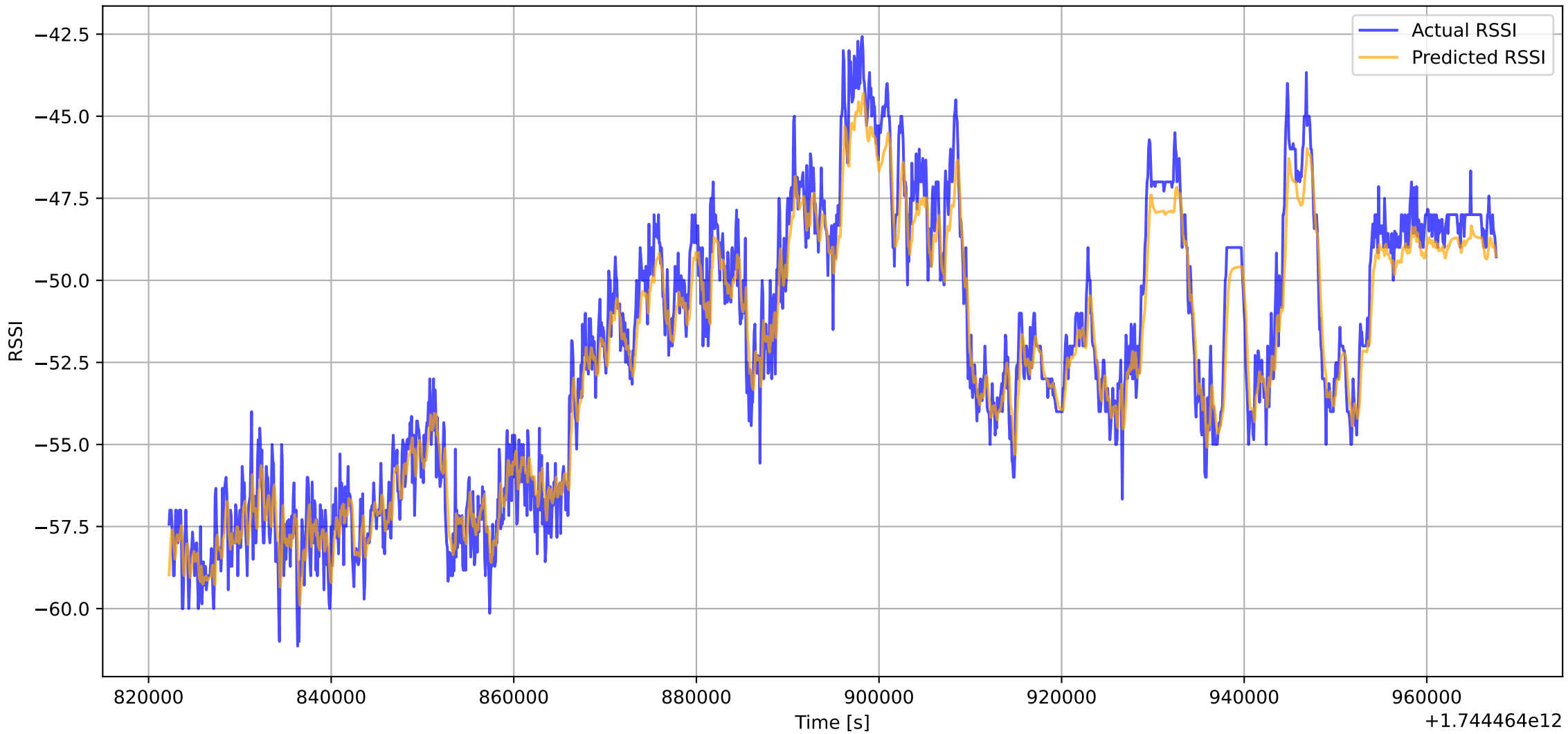
1744397085760
1744397085841
1744397086014
1744397086103
1744397086172

...

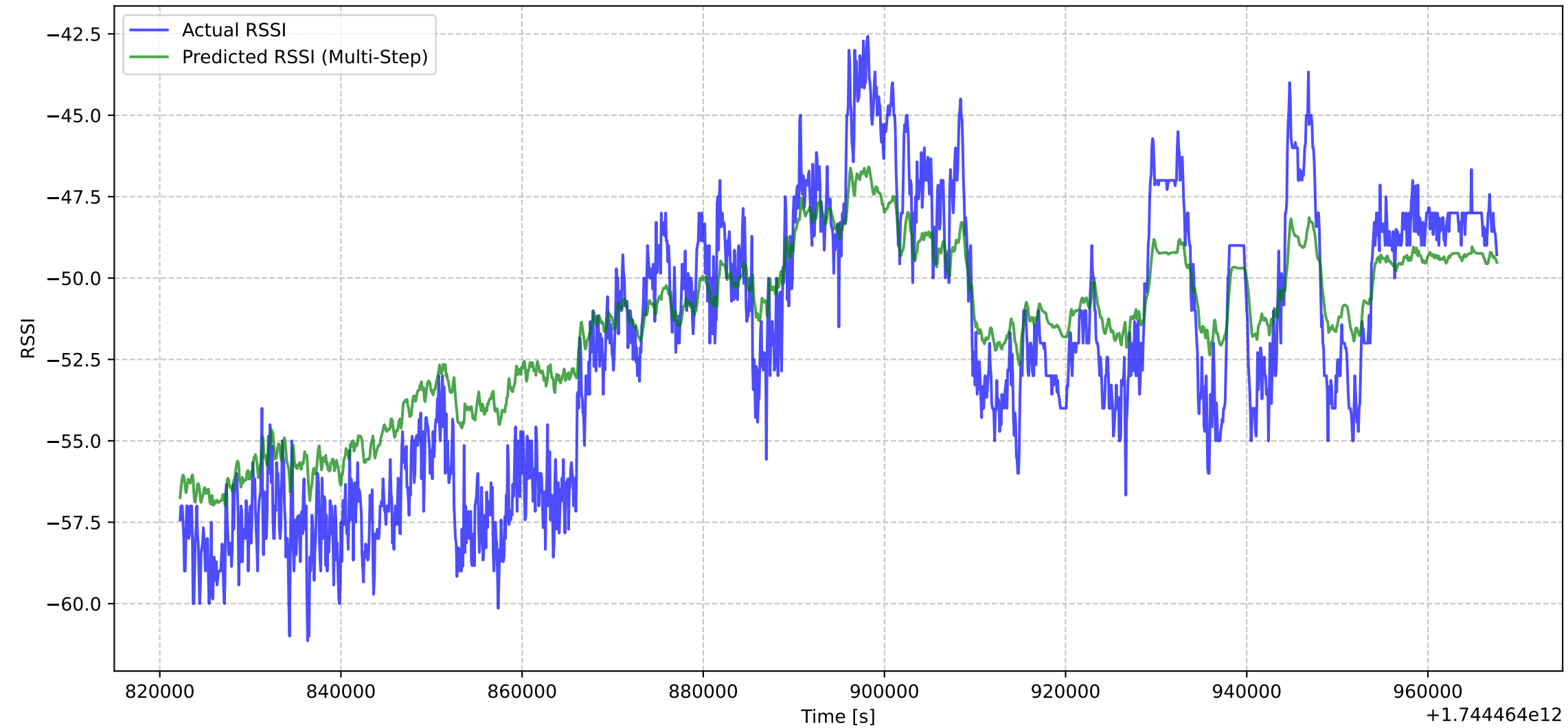
Last 5 timestamps AFTER:

1744464971041
1744464971124
1744464971209
1744464971281
1744464971366

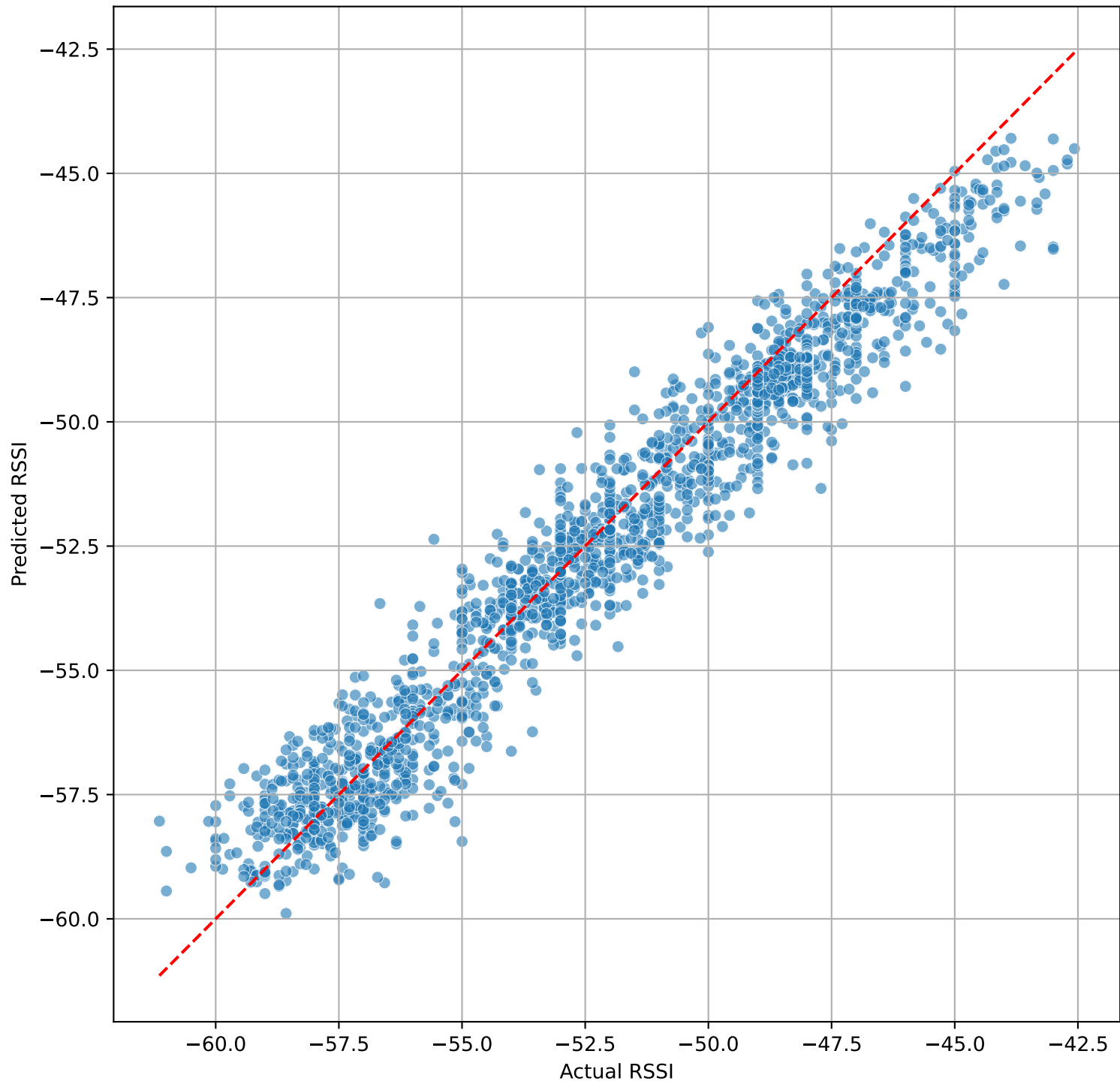
Actual vs Predicted RSSI (First Step)



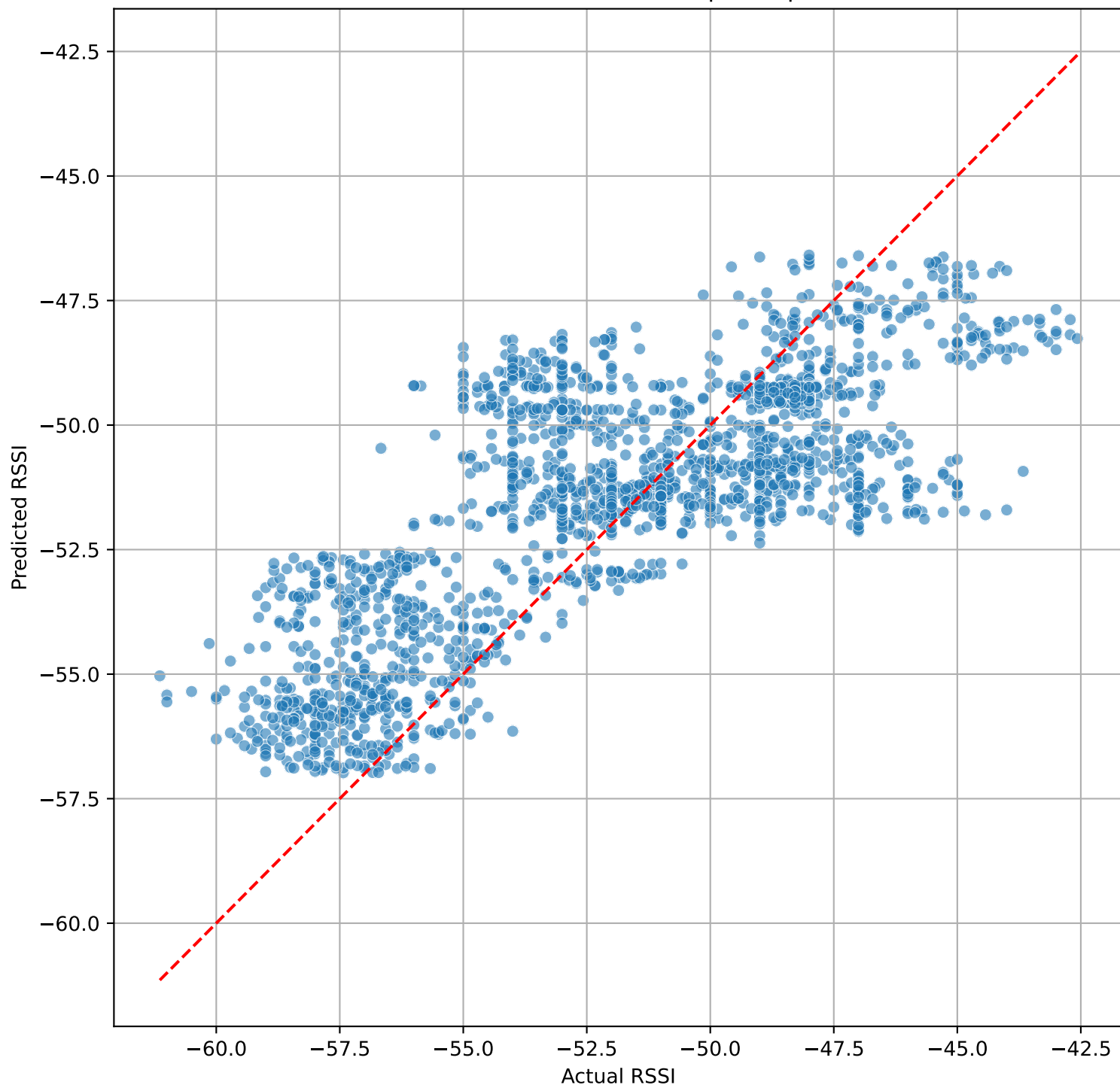
Actual vs Predicted RSSI (Multi-Step (step 48 $\approx$ 4.0 s))



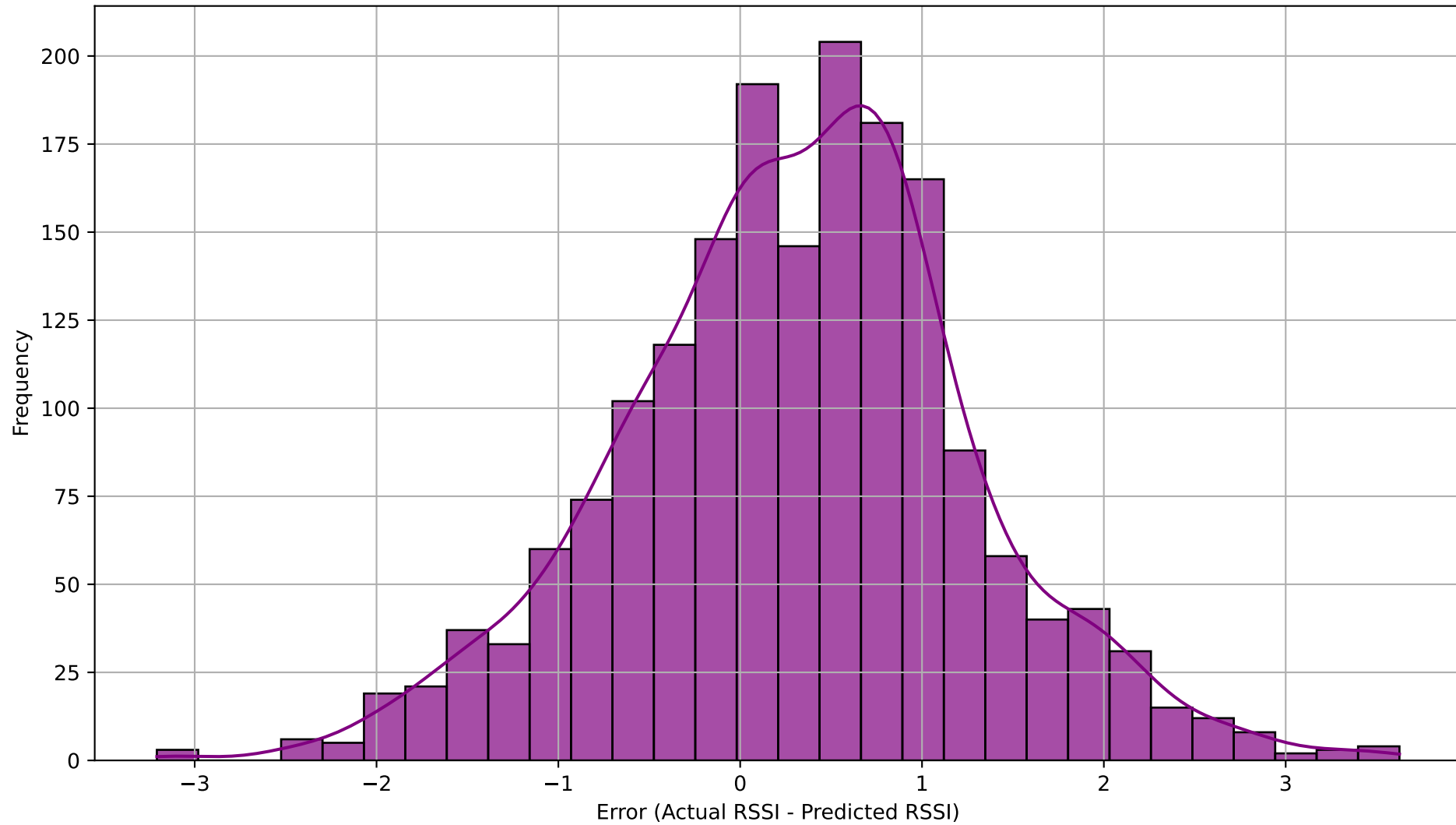
Actual vs Predicted RSSI (First Step)



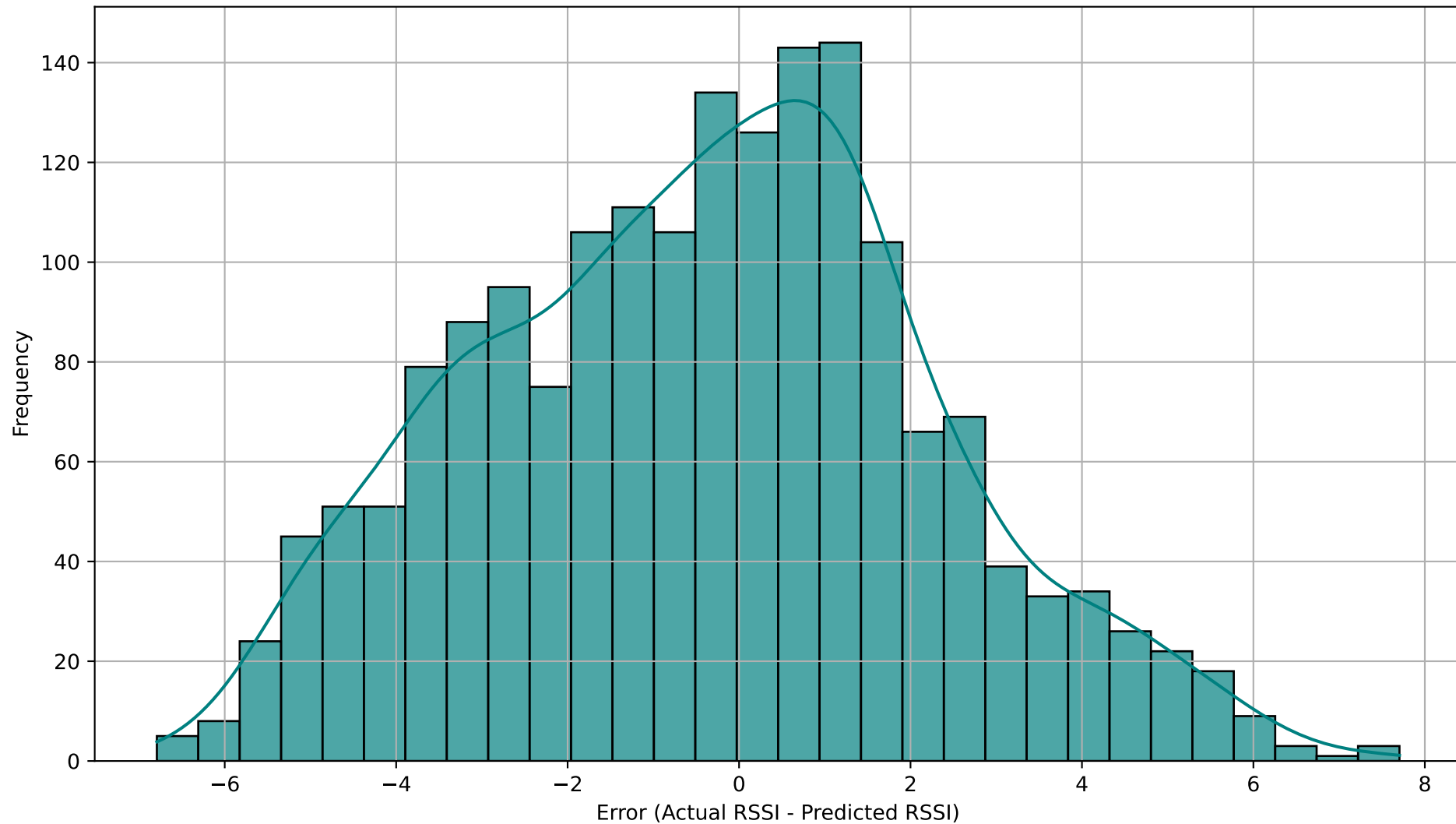
Actual vs Predicted RSSI (Last Step - step 48 $\approx$ 4.0 s)



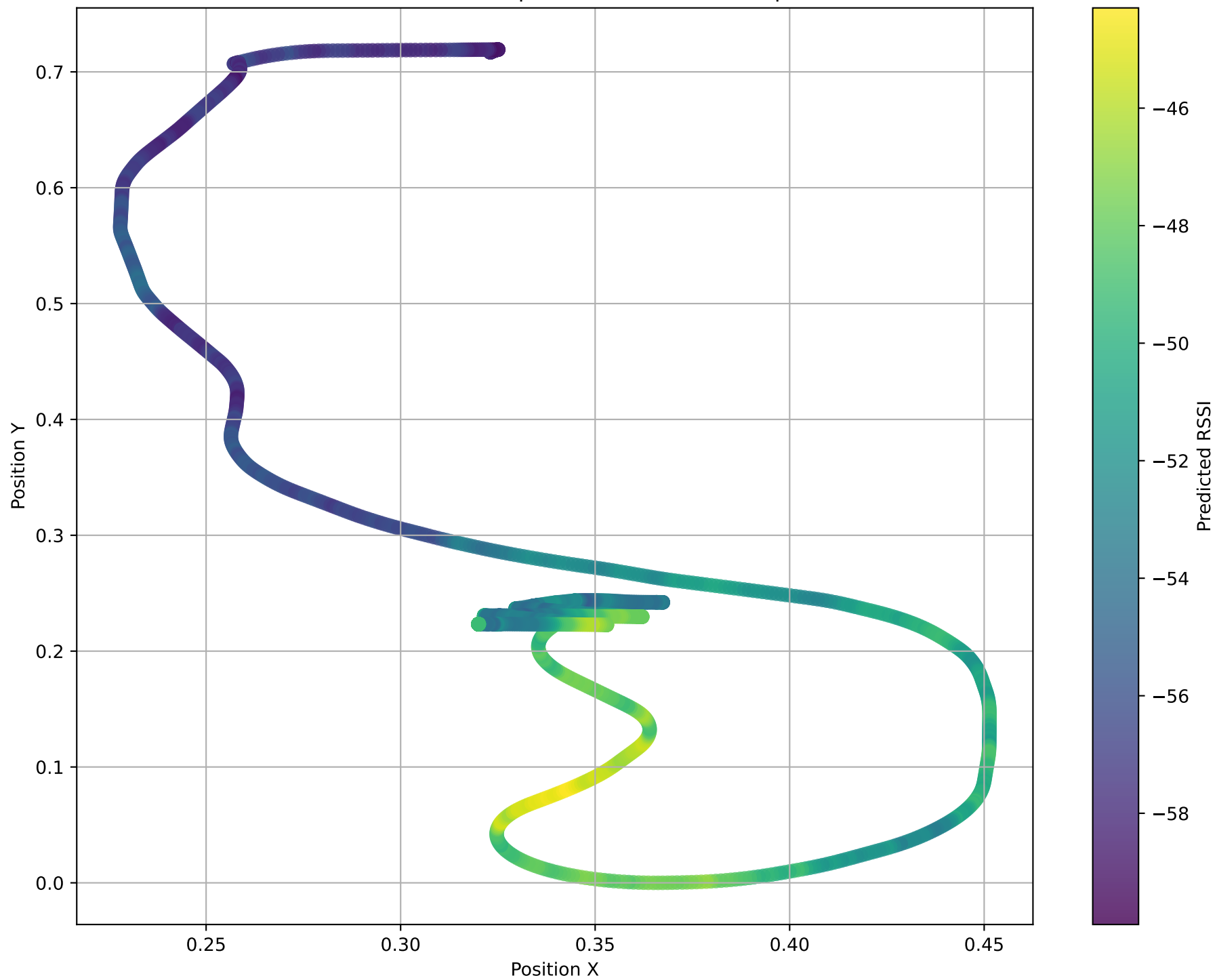
Distribution of Prediction Errors (First Step)



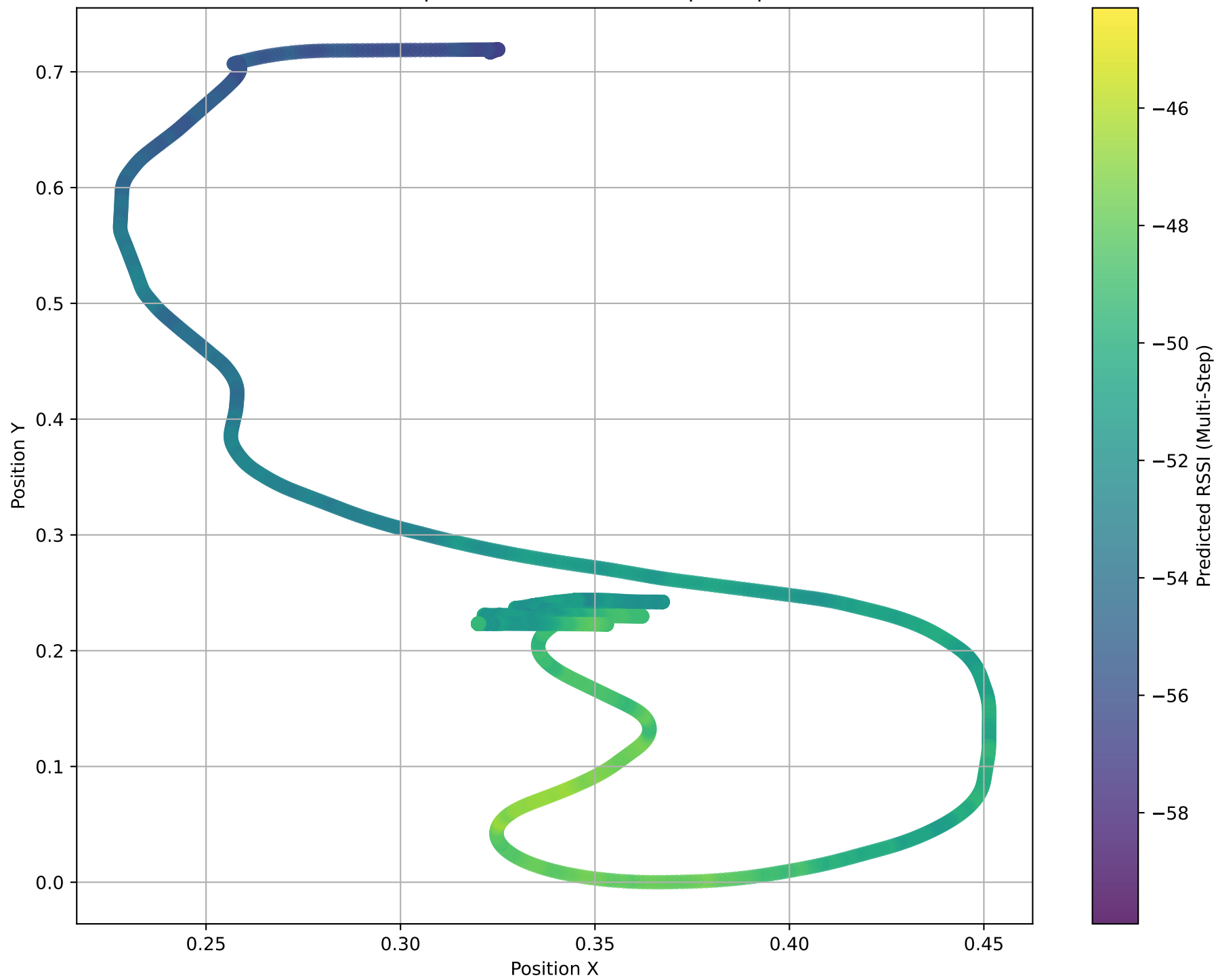
Distribution of Prediction Errors (Last Step - step 48 $\approx$ 4.0 s)



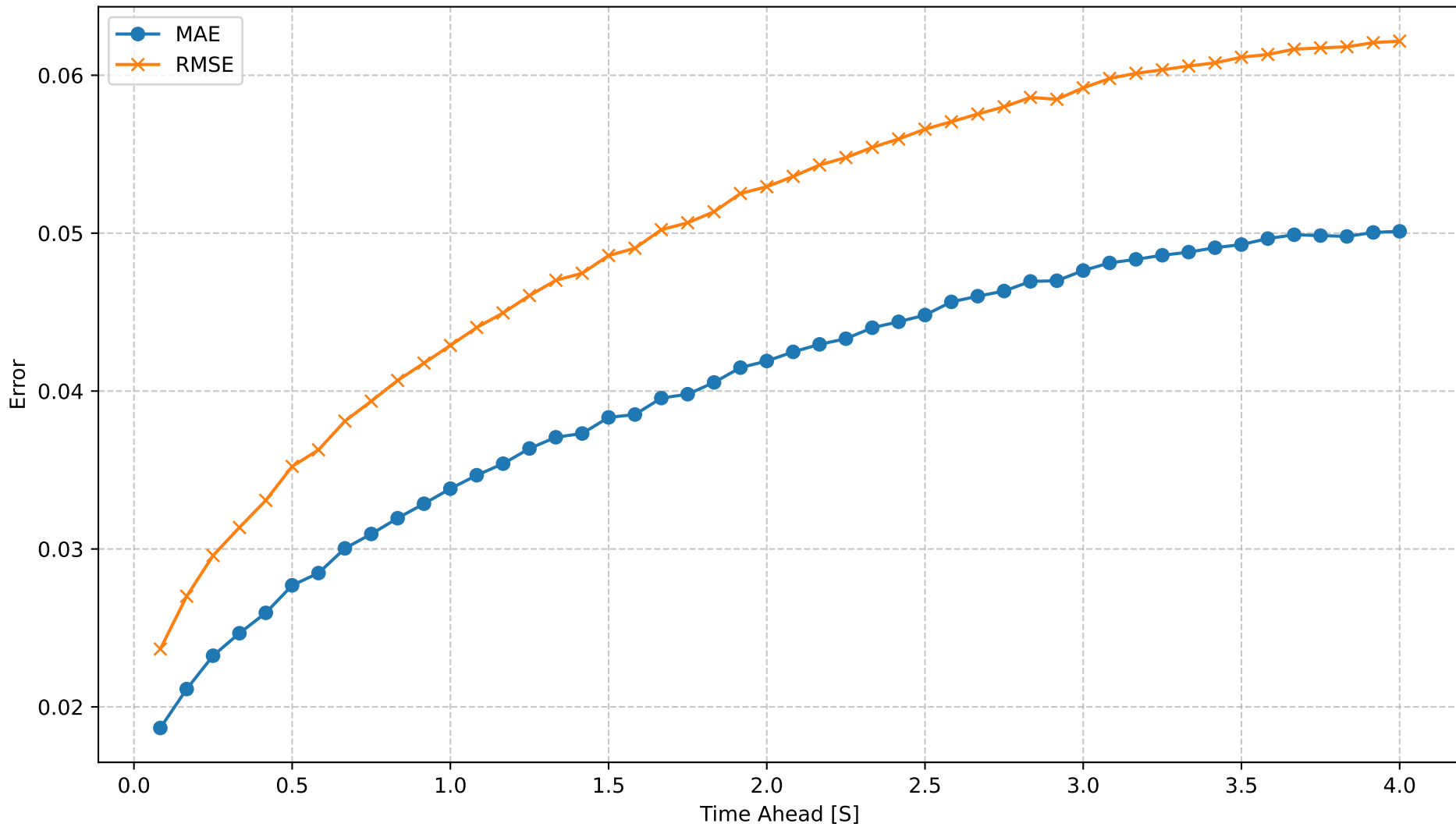
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 48 $\approx$ 4.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_ajust.parquet
Modelo: LSTM
Forecast steps: 48

MAE / RMSE (dB)

- 1º passo : 0.02 / 0.02
- Mediana : 0.04 / 0.05
- 48º passo : 0.05 / 0.06

Crescimento do erro: 168.5% → Degrada Rápido (> 50 %)

3 piores passos (MAE): 48 (0.05), 47 (0.05), 44 (0.05)

Recomendação:

Considere aumentar o histórico de entrada ou usar um modelo mais profundo para melhorar multi-step.